

Stage-2 Hybrid TEC-AO Feedback Analysis

Fixed Model Failure, Improved Attended Run, and Control Strategy for Future Experiments

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1 Purpose of this report

This report merges the analysis of three related components of the Stage-2 spectrograph stabilisation work:

1. the general modelling argument comparing **adaptive warm-up-derived models** against a **fixed model**;
2. the **20 April fixed-model run**, which initially stabilised but later failed under environmental/OPL drift;
3. the **22–24 April attended run**, which showed improved long-duration behaviour, recovery from a mid-run excursion, and clearer limitations of AO travel management.

The goal is to present the numerical evidence, figures, and control interpretation in one coherent document suitable for discussion with supervisors and later adaptation into a paper/report section.

Core message: the system sensitivities are not constant between experiments. A purely fixed model can become inaccurate or even wrong in sign. The most defensible strategy is therefore **adaptive warm-up modelling + TEC-primary control + AO fine trim**, with the fixed model used only as a weak prior.

2 Background: adaptive versus fixed modelling

The spectrograph response can be approximated locally using linear sensitivity models:

$$\Delta X_k = \alpha_T \Delta T_k + \alpha_P \Delta P_k + \alpha_H \Delta H_k + \eta_k, \quad (1)$$

$$\Delta Y_k = \beta_T \Delta T_k + \beta_P \Delta P_k + \beta_H \Delta H_k + \varepsilon_k, \quad (2)$$

$$\Delta \text{OPL}_k = g_T \Delta T_k + g_P \Delta P_k + g_H \Delta H_k + \xi_k. \quad (3)$$

Here ΔT_k , ΔP_k , and ΔH_k represent thermal, pressure, and humidity/water-vapour changes, while α , β , and g are sensitivity coefficients.

2.1 Fixed model assumption

A fixed model assumes that the coefficients

$$\alpha_T, \alpha_P, \alpha_H, \beta_T, \beta_P, \beta_H, g_T, g_P, g_H$$

remain valid across different days and different operating states. This would only be reasonable if the plant were close to time-invariant and the thermal/environmental coupling were repeatable.

Main problem: the experimental data show that the sensitivities change substantially between runs. In some cases even the **sign** changes, so a fixed controller can push in the wrong direction.

2.2 Adaptive warm-up model

Instead, the model coefficients can be estimated during the warm-up of each individual run:

$$\hat{\theta}_{\text{warmup}} = \arg \min_{\theta} \sum_{k \in \text{warmup}} (y_k - X_k \theta)^2 + \lambda \|\theta\|^2. \quad (4)$$

This gives run-specific coefficients for the current plant state.

Important point: the adaptive model does not assume that yesterday’s sensitivity is still valid today. It identifies the present response from the current run itself.

3 Earlier three-experiment comparison

The earlier comparison used three runs:

- **Experiment A:** `rt_stage2_hybrid_temp_ao_23_03_26.csv`
- **Experiment B:** `rt_stage2_final_v6.csv`
- **Experiment C:** `rt_stage2_20260415.fixed_model_hybrid_analysis_ready.csv`

3.1 Performance summary

Experiment	RMS dX (px)	RMS dY (px)	RMS drift (px)	Mean $ dY $ (px)	Frac($ dY \leq 0.10$)	OPL std (μm)
A	0.751	0.371	0.838	0.231	0.390	1.734
B	0.249	0.523	0.580	0.164	0.176	0.905
C	0.301	4.982	4.991	3.301	0.064	5.820

Table 1: Main comparison metrics across the earlier three experiments.

Highs:

- **Experiment A** gave the best centroid/image stability.
- **Experiment B** gave the calmest thermal/OPL behaviour.

Low:

- **Experiment C** was the worst overall, with very large dY , drift, OPL, and TEC variability.

3.2 Evidence that sensitivities change between runs

The strongest evidence comes from Experiment C, where the warm-up-fitted sensitivities differed strongly from the fixed model values.

Quantity	Warm-up fitted value	Fixed value used	Difference
g_T for OPL ($\mu\text{m}/^\circ\text{C}$)	-11.961	-4.100	-7.861
g_P for OPL ($\mu\text{m}/\text{hPa}$)	-6.773	+0.250	-7.023
g_H for OPL ($\mu\text{m}/(\text{g m}^{-3})$)	-0.390	-0.670	+0.280
β_T for dY (px/ $^\circ\text{C}$)	-4.033	+1.605	-5.638
β_P for dY (px/hPa)	-3.290	-0.633	-2.656
β_H for dY (px/(g m $^{-3}$))	-0.097	+0.742	-0.839

Table 2: Experiment C: warm-up-derived sensitivities compared with the fixed coefficients.

Critical result: these are not small calibration drifts. Several coefficients differ substantially and some change sign. If the sign is wrong, the controller may push in the wrong direction.

The comparison between warm-up fitted sensitivities and logged early adaptive/live coefficients also showed mismatch across experiments. For example:

- **Experiment A:** warm-up OPL temperature sensitivity $\approx +9.08 \mu\text{m}/^\circ\text{C}$, while logged early adaptive value $\approx +0.94 \mu\text{m}/^\circ\text{C}$.
- **Experiment B:** warm-up OPL temperature sensitivity $\approx +0.66 \mu\text{m}/^\circ\text{C}$, while logged early value $\approx -4.10 \mu\text{m}/^\circ\text{C}$.
- **Experiment C:** warm-up OPL temperature sensitivity $\approx -11.96 \mu\text{m}/^\circ\text{C}$, while logged early value $\approx -4.10 \mu\text{m}/^\circ\text{C}$.

Correct interpretation:

1. use warm-up identification to capture the current run;
2. use the adaptive result as the main model;
3. use the fixed model only as a weak prior;
4. keep the empirical image error as the main stabilising anchor.

4 20 April fixed-model experiment

4.1 Objective

The 20 April experiment evaluated a fixed-model TEC feedback controller applied to spectrograph stabilisation. The controller used pre-derived coefficients to predict and correct optical path length and centroid drift.

The key questions were:

- Can the fixed model stabilise dY ?
- Does TEC maintain authority under environmental drift?
- Does AO act as a fine stabiliser or become involved in large-drift behaviour?

4.2 Dataset and overall metrics

Metric	Value
Duration	36.95 h
RMS dX	0.4394 px
RMS dY	5.8756 px
RMS radial error	5.8920 px
Fraction $ err \leq 0.1$ px	9.83%
Fraction $ err \leq 0.2$ px	17.74%
Fraction $ err \leq 0.5$ px	28.95%
TEC actions	464
AO actions	38

Table 3: Overall performance of the 20 April fixed-model run.

4.3 Full system behaviour



Figure 1: 20 April fixed-model run: full time-series showing pixel drift, OPL variation, TEC control, AO response, and environmental parameters. Highlighted regions show behavioural phases; yellow bands indicate suspected environmental/HVAC-like disturbances.

4.4 Behavioural windows

The run showed four main regimes:

1. stable hybrid plateau;
2. breakaway;
3. bad negative plateau;
4. failed recovery.

Window	Time range (h)	RMS dY (px)	RMS err (px)	Mean dY (px)	OPL change (μm)	TEC/AO actions
Stable plateau	5.01–14.48	0.3801	0.3949	-0.1447	+0.8558	30 / 33
Breakaway	14.52–20.47	6.6485	6.6793	-5.8704	-10.1123	19 / 5
Bad negative plateau	20.51–28.08	10.1119	10.1328	-10.1107	+1.0174	200 / 0
Failed recovery	28.13–36.95	4.9288	4.9434	-2.8499	-4.3133	208 / 0

Table 4: Event-window analysis of the 20 April fixed-model run.

Positive result: the fixed model initially produced a stable plateau with RMS $dY = 0.3801$ px and RMS radial error = 0.3949 px.

Failure mode: after the breakaway, the system entered a sustained negative plateau near $dY \approx -10$ px. TEC continued to act but did not recover the centroid.

4.5 TEC command to pixel-response lag

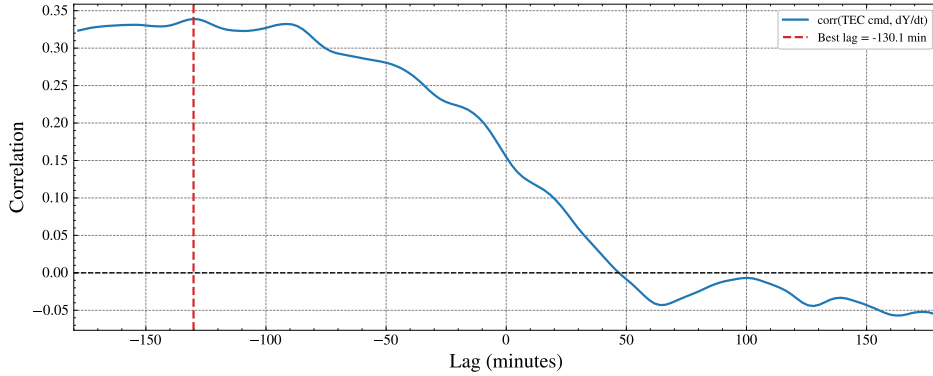


Figure 2: 20 April run: lag correlation between TEC command and pixel-response derivative.

The lag analysis gave:

- best lag: **-130.1 min**;
- correlation at best lag: **0.339**.

Interpretation: the negative lag suggests that pixel drift was already changing before the TEC correction. The controller was reacting to an environmental/OPL-driven drift rather than predicting it.

4.6 Interpretation of the 20 April run

The 20 April run demonstrates that a fixed model can work during calm/stable environmental conditions, but it is not robust to drift outside the training/assumed operating regime. The breakaway and negative plateau show that the TEC loop was reactive and could settle at the wrong centroid offset. AO also approached a large travel offset, indicating that AO was being pulled into a problem that should have remained primarily thermal/OPL controlled.

Main conclusion from 20 April: the controller did not fail because feedback is impossible. It failed because the fixed model was not sufficiently predictive or adaptive under changing environmental and OPL conditions.

5 22–24 April attended hybrid experiment

5.1 Overview

The 22–24 April attended run tested an improved TEC-primary controller with AO-assisted stabilisation. The experiment ran over approximately 50 hours and included intentional restart/recovery phases.

5.2 Dataset summary

Metric	Value
Start time	22 Apr 2026, 09:10
End time	24 Apr 2026, 10:48
Total duration	49.63 h
Valid science rows	1238
TEC actions	262
AO actions	109

Table 5: Dataset summary for the 22–24 April attended run.

5.3 Full-run performance

Metric	Value
RMS dX	0.5254 px
RMS dY	1.2015 px
RMS radial error	1.3113 px
Fraction $ err \leq 0.1$ px	8.24%
Fraction $ err \leq 0.2$ px	20.68%
Fraction $ err \leq 0.5$ px	47.82%

Table 6: Full-run performance metrics for the 22–24 April attended experiment.

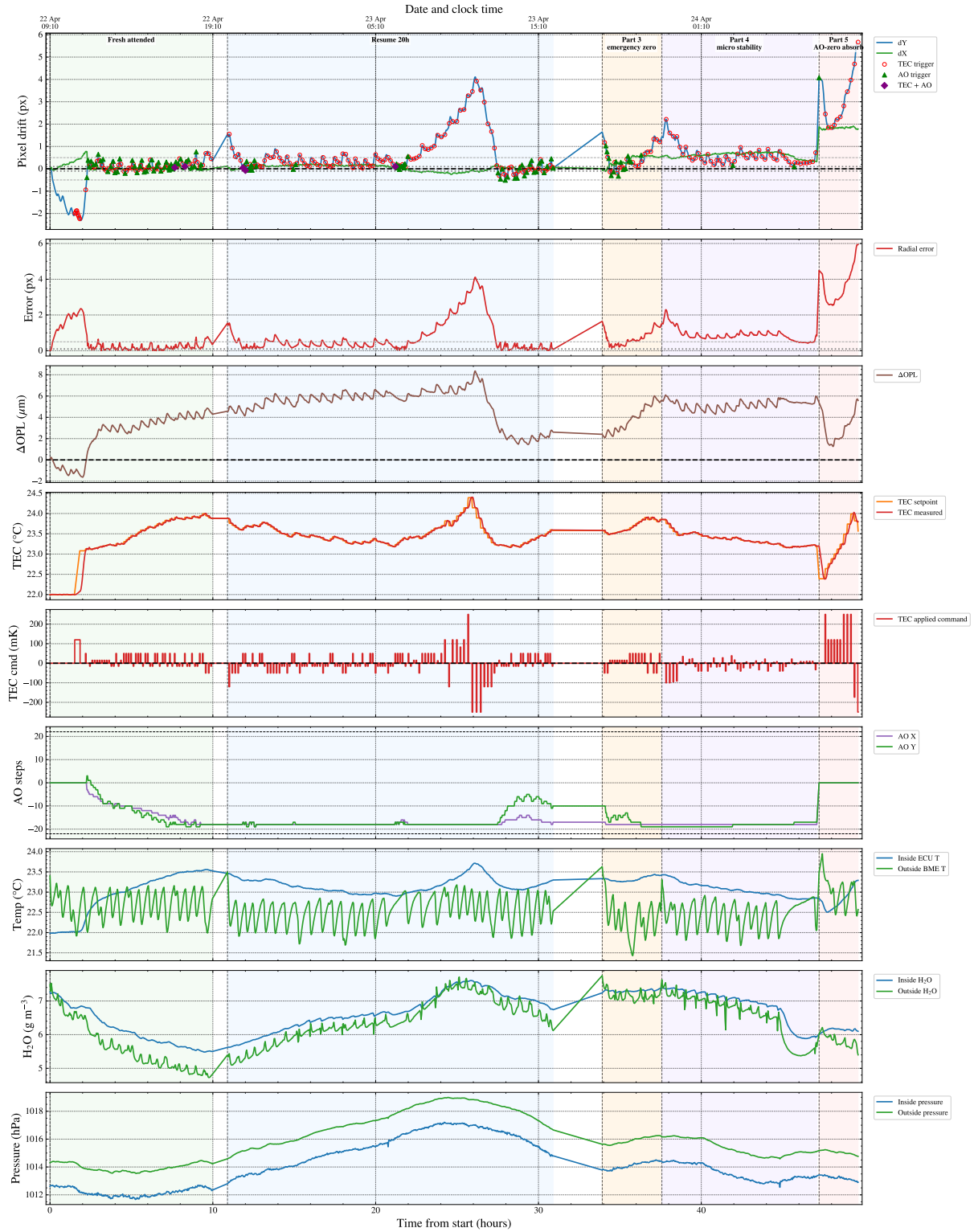


Figure 3: 22–24 April attended run: full multi-panel behaviour showing pixel drift, radial error, OPL, TEC setpoint/measured temperature, TEC commands, AO cumulative steps, and environmental channels.

5.4 Performance excluding final AO-zero absorb section

The final Part 5 AO-zero/TEC absorb section was a deliberate recentering/absorb experiment and should not be treated as nominal stability data.

Metric	Value
Duration	47.10 h
RMS dX	0.3379 px
RMS dY	0.9778 px
RMS radial error	1.0345 px
Fraction $ err \leq 0.1$ px	8.67%
Fraction $ err \leq 0.2$ px	21.77%
Fraction $ err \leq 0.5$ px	50.34%

Table 7: 22–24 April performance excluding the deliberate final AO-zero absorb/recentering section.

5.5 Phase-based performance

Phase	Duration (h)	RMS dX (px)	RMS dY (px)	RMS err (px)
Fresh attended	9.97	0.1778	0.8131	0.8323
Resume 20h	20.00	0.1322	1.1826	1.1900
Part 3 emergency zero	3.60	0.4294	0.7178	0.8364
Part 4 micro-stability	9.52	0.6159	0.7078	0.9382
Part 5 AO-zero absorb	2.40	1.8294	3.2696	3.7466

Table 8: RunLabel-based phase performance in the 22–24 April attended run.

5.6 Trimmed feedback-zone performance

A trimmed presentation dataset retained the useful warm-up/stable-feedback/recovered-feedback/micro-stability regions while excluding the large deliberate excursions.

Metric	Value
Rows	886
Clock coverage	22 Apr 09:10 – 24 Apr 08:15
RMS dX	0.3569 px
RMS dY	0.6467 px
RMS radial error	0.7386 px
Fraction $ err \leq 0.1$ px	9.48%
Fraction $ err \leq 0.2$ px	24.60%
Fraction $ err \leq 0.5$ px	56.66%

Table 9: Trimmed feedback-zone performance for the 22–24 April attended run.

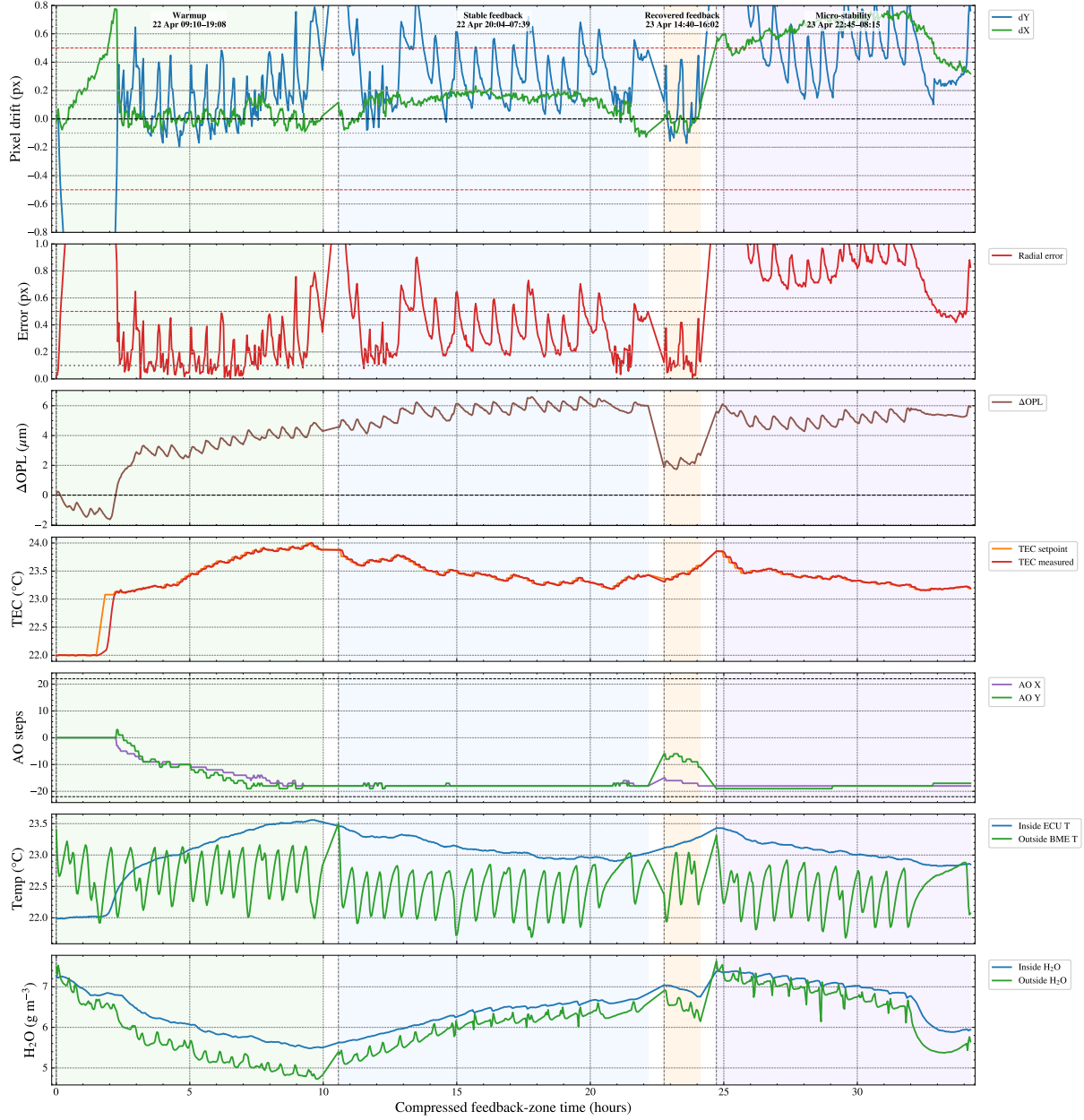


Figure 4: 22–24 April: trimmed feedback-zone plot focusing on useful control regions and sub-pixel behaviour.

Practical reading: the full run gives the honest overall number, but the trimmed feedback-zone result better represents useful controller performance. In the trimmed control regions, RMS dY was 0.6467 px and 56.66% of points were within 0.5 px radial error.

6 Triangle excursion in the 22–24 April run

6.1 Observation

A smooth triangular dY excursion occurred during the Resume 20h section. The event occurred approximately over:

$$22.5 \leq t_{\text{hr}} \leq 29.5,$$

corresponding approximately to 23 Apr 07:42–14:38. The dY value rose to a peak near 4 px and then recovered close to zero by approximately 15:10 on 23 April.

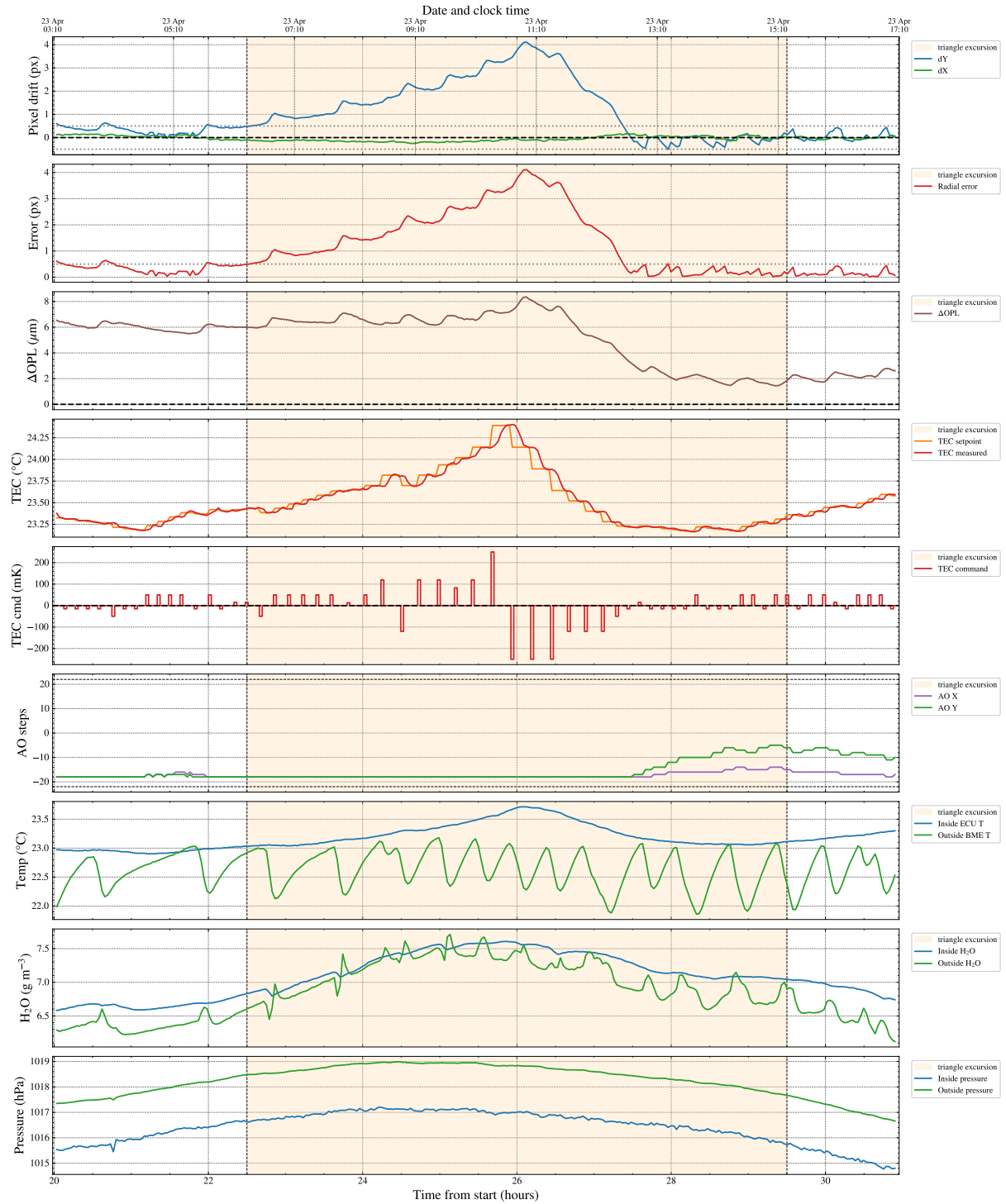


Figure 5: 22–24 April: zoomed diagnostic plot of the triangular dY excursion.

6.2 Window statistics

Window	Duration (h)	RMS dX (px)	RMS dY (px)	RMS err (px)	AO motion
Pre-triangle stable control	11.60	0.1405	0.4331	0.4553	$(-18, -18) \rightarrow (-18, -18)$
Triangle excursion	6.93	0.1295	1.9232	1.9276	$(-18, -18) \rightarrow (-15, -6)$
Post-triangle recovery	1.40	0.0479	0.1867	0.1927	$(-15, -6) \rightarrow (-17, -10)$
Micro-stability Part 4	9.52	0.6159	0.7078	0.9382	$(-18, -19) \rightarrow (-18, -17)$

Table 10: Selected event-window performance for the 22–24 April attended run.

6.3 Correlation analysis during the triangle

During the triangular excursion, dY correlated most strongly with internal temperature, TEC measured temperature, TEC setpoint, OPL, humidity, and pressure.

Variable	Correlation with dY
ECU/internal temperature	0.919
TEC measured temperature	0.913
TEC setpoint	0.865
Δ OPL	0.848
Inside H ₂ O	0.802
Outside H ₂ O	0.789
Outside pressure	0.742
Inside pressure	0.706
AO cumulative Y	-0.645
AO cumulative X	-0.604
TEC command	-0.112
Outside BME temperature	0.094

Table 11: Correlation with dY during the triangular excursion.

Interpretation: the event was not primarily a direct outside-temperature event. It was most strongly coupled to internal temperature, TEC temperature, OPL, and water-vapour/pressure changes.

6.4 Rise and recovery mechanism

During the triangle rise:

Quantity	Start	Peak	Change
dY	0.5344 px	4.1104 px	+3.5759 px
dX	-0.0918 px	-0.1128 px	-0.0210 px
Radial error	0.5423 px	4.1119 px	+3.5697 px
ΔOPL	6.2384 μm	8.3477 μm	+2.1093 μm
TEC setpoint	23.4200°C	24.1406°C	+0.7206°C
TEC measured	23.3739°C	24.1527°C	+0.7787°C
ECU temperature	22.9890°C	23.7130°C	+0.7240°C
ECU H_2O	6.6954 g m ⁻³	7.5621 g m ⁻³	+0.8667 g m ⁻³
ECU pressure	1016.43 hPa	1017.03 hPa	+0.60 hPa
BME temperature	22.22°C	22.28°C	+0.06°C
BME H_2O	6.4770 g m ⁻³	7.4499 g m ⁻³	+0.9729 g m ⁻³
BME pressure	1018.20 hPa	1018.82 hPa	+0.62 hPa
AO cumulative position	(-18, -18)	(-18, -18)	no motion

Table 12: Triangle rise: changes from start to dY peak.

During the recovery/fall:

Quantity	Start	End	Change
dY	4.0249 px	0.0308 px	-3.9941 px
dX	-0.1069 px	0.0644 px	+0.1714 px
Radial error	4.0263 px	0.0714 px	-3.9549 px
ΔOPL	8.1672 μm	2.6087 μm	-5.5584 μm
TEC setpoint	24.1406°C	23.5806°C	-0.5600°C
TEC measured	24.1371°C	23.5933°C	-0.5438°C
ECU temperature	23.7050°C	23.3000°C	-0.4050°C
ECU H_2O	7.5648 g m ⁻³	6.7400 g m ⁻³	-0.8248 g m ⁻³
ECU pressure	1017.00 hPa	1014.80 hPa	-2.20 hPa
BME H_2O	7.3315 g m ⁻³	6.1194 g m ⁻³	-1.2121 g m ⁻³
BME pressure	1018.82 hPa	1016.66 hPa	-2.16 hPa
AO cumulative position	(-18, -18)	(-17, -10)	AO Y unloaded by +8 steps

Table 13: Triangle recovery/fall: changes from peak region to recovery.

6.5 Physical explanation of the triangle

The triangle was not caused by AO motion during the rise, since AO remained fixed at (-18, -18). Instead, AO was blocked or unavailable, and the controller relied on TEC.

Layman explanation: TEC is the slow but powerful steering wheel; AO is the fast but limited fine knob. During the triangle rise, the fine knob was already near its end stop, so the controller used the big steering wheel. The TEC temperature increase moved the optical path and shifted the spectrum in Y. Later, when the controller reversed and AO regained some usable range, the system returned near zero.

In more technical terms:

- AO was near its negative travel boundary and did not move during the rise.
- The controller applied a net positive TEC response of approximately +0.77°C.
- TEC/internal temperature increased, OPL increased, and dY increased to approximately 4.11 px.

- Once TEC reversed and AO began unloading, dY returned close to zero by about 15:10 on 23 April.

Control issue identified: if AO is saturated, TEC must not be allowed to make a large recentering move without predictive damping and sign/authority checks. Otherwise, the TEC can create a controlled but large thermal excursion.

7 Direct comparison: 20 April vs 22–24 April

Metric	20 April fixed-model run	22–24 April attended run
Duration	36.95 h	49.63 h
RMS dY full run	5.8756 px	1.2015 px
RMS radial error full run	5.8920 px	1.3113 px
Fraction $ err \leq 0.1$ px	9.83%	8.24%
Fraction $ err \leq 0.2$ px	17.74%	20.68%
Fraction $ err \leq 0.5$ px	28.95%	47.82%
TEC actions	464	262
AO actions	38	109
Worst behaviour	sustained -10 to -13 px runaway	recovered triangle + final deliberate AO reset
Nominal/control RMS dY	0.3801 px initially; later failed	0.6467 px trimmed feedback-zone
Recovery quality	poor; failed recovery	recovered near zero after excursion

Table 14: Direct comparison between the 20 April fixed-model run and the 22–24 April attended run.

Main improvement: the 22–24 April run avoided the sustained -10 to -13 px runaway seen on 20 April and recovered from the main triangular excursion.

Remaining limitation: the 22–24 April run still showed ripple, AO travel-limit issues, and a TEC-driven triangular excursion when AO was saturated.

8 Control strategy implications

8.1 Empirical correction

The empirical temperature correction from measured image drift is:

$$e_T^{(\text{emp})} = -\frac{\Delta Y}{\hat{\beta}_{Y,T}}. \quad (5)$$

This is useful because it directly ties the temperature correction to the measured image motion.

8.2 Model-based correction

A model-based correction can be estimated using both pixel and OPL information:

$$e_T^{(\text{px})} = f_{\text{pixel}}^{-1}(\Delta X, \Delta Y, \Delta P, \Delta H), \quad (6)$$

$$e_T^{(\text{opl})} = -\frac{\Delta \text{OPL}}{\hat{g}_T}. \quad (7)$$

These can be combined as:

$$e_T^{(\text{model})} = \frac{1}{2} \left(e_T^{(\text{px})} + e_T^{(\text{opl})} \right). \quad (8)$$

8.3 Confidence-weighted fusion

The final correction estimate may be written as:

$$e_T^{(\text{fused})} = (1 - w_m - w_f) e_T^{(\text{emp})} + w_m e_T^{(\text{model})} + w_f e_T^{(\text{fixed})}, \quad (9)$$

where w_m depends on model confidence and w_f is small.

Meaning: the controller should not blindly trust the model. It should remain anchored to measured image error, while using OPL/environment models as supporting information.

8.4 TEC control law

A conservative event-based TEC control law can be written as:

$$\Delta u_k = \text{sat} \left(K_P e_k + K_I \sum_{j=1}^k e_j + K_D \dot{e}_k \right). \quad (10)$$

The controller should include:

- step caps;
- holdoff/settle windows;
- reversal freeze;
- persistence requirement;
- calm gating;
- predictive damping during AO-saturated states.

8.5 AO role

AO should not be the primary stabiliser. Its role should be:

- fine trim only;
- only when TEC is calm;
- only when residual error is modest;
- with active monitoring of cumulative travel.

The AO step estimate is:

$$\Delta s = M^{-1} \begin{bmatrix} -\Delta X \\ -\Delta Y \end{bmatrix}, \quad (11)$$

where M is the AO calibration matrix.

Correct AO philosophy: AO should trim the residual, not rescue a badly unstable thermal loop.

9 Advantages and limitations of regression-based adaptive modelling

9.1 Advantages

- simple and interpretable;
- fast to compute;
- suitable for warm-up data windows;
- physically understandable;
- easy to explain in supervisor discussions and reports.

9.2 Limitations

- assumes approximate local linearity;
- can be affected by collinearity;
- may overfit if warm-up data are poor;
- does not naturally capture lag, hysteresis, or hidden thermal state.

Practical conclusion:

adaptive warm-up regression + confidence gating + TEC-primary control + AO fine trim

10 Final conclusion

The experiments together show a clear progression.

The earlier A/B/C comparison established that fixed coefficients are not reliable across runs. Experiment A showed that good image stability is achievable; Experiment B showed that calmer OPL/thermal behaviour is possible; Experiment C showed that mismatched fixed coefficients and unstable control logic can drive large drift.

The 20 April run then demonstrated this failure mode directly in the Stage-2 hybrid system. It initially reached a stable plateau with RMS $dY = 0.3801$ px, but later entered a large breakaway and bad negative plateau near $dY \approx -10$ px. Lag analysis indicated that the controller was reacting to drift that had already begun.

The 22–24 April attended run showed a major improvement. Full-run RMS dY fell to 1.2015 px, and the trimmed useful feedback-zone RMS dY was 0.6467 px. The system did not suffer the sustained -10 to -13 px runaway seen on 20 April. It also recovered from a triangular mid-run excursion. However, that event revealed a remaining limitation: when AO was saturated, TEC applied a large thermal move, producing a temporary OPL and dY excursion.

Final engineering conclusion: the next controller should use an adaptive warm-up-derived model, TEC as the primary actuator, and AO only as fine trim. It must also manage AO travel proactively and damp TEC moves when AO is saturated.